

## **Milestone 2 – Data Selection and Project Proposal**

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# Project Overview and Initial Plan Review

## A. Introduction

The objective of this project is to apply advanced predictive analytics techniques to a real-world problem: **Detecting credit card fraud**. Our goal is to build, evaluate, and optimize a machine learning model that accurately predicts whether a transaction is **fraudulent (Class=1)** or **legitimate (Class=0)**. This project will focus heavily on managing the class imbalance inherent in fraud data and using appropriate metrics to deliver a valuable predictive solution.

Our initial high-level plan involves four phases:

1. Data Acquisition and Project Definition
2. Data Cleaning and Exploration
3. Model Development and Optimization
4. Evaluation, Documentation, and Final Submission.

This proposal formalizes the first two phases and outlines the specific methodology for the predictive modeling stage.

## B. Initial Dataset Selection and Data Dictionary

For our predictive analytics project, we have selected the Credit Card Fraud Detection Dataset from Kaggle, made publicly available by European cardholders.

- Dataset Name: Credit Card Fraud Detection
- Source: Kaggle
- Access URL: <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>
- Key Characteristics: The dataset contains 284,807 transactions that occurred over two days in September 2013. Critically, the data is highly unbalanced, with the fraudulent class accounting for only 0.172% of all transactions (492 instances of fraud).
- Data Dictionary:
  - Features V1-V28: Principal Component Analysis (PCA)-transformed features are used due to confidentiality concerns. These features are numerical and non-interpretable in their raw form.
  - Time: Seconds elapsed between each transaction and the first transaction in the dataset.
  - Amount: The transaction amount.
  - Class (Target Variable): Binary label, where 1 indicates a fraudulent transaction, and 0 indicates a legitimate transaction.

This dataset presents an ideal challenge for predictive analytics, necessitating robust methods to identify rare events accurately.

## 2. Methodology and Evaluation

### A. Proposed Model(s) and Rationale

We plan to use a tiered approach, starting with a simple model for a performance baseline and progressing to more complex ensemble methods best suited for imbalanced classification problems:

1. Baseline Model: Logistic Regression
  - Rationale:** A simple linear model will establish a computationally efficient and easily interpretable baseline score against which all subsequent, complex models will be measured.
2. Primary Model: Random Forest Classifier
  - Rationale:** Random Forest is an ensemble method known for its robustness against overfitting and its ability to handle large, high-dimensional datasets without requiring extensive feature scaling. It will serve as our primary model for optimization.
3. Advanced Model: XGBoost (Extreme Gradient Boosting)
  - Rationale:** Gradient boosting algorithms frequently yield state-of-the-art results in classification. XGBoost's ability to handle weighted classes and its high predictive power make it essential for effectively classifying the highly underrepresented cases of fraud.

Before training these models, we will apply sampling techniques (e.g., SMOTE or ADASYN oversampling of the minority class, or targeted undersampling of the majority class) to the training data to mitigate the class imbalance bias.

### B. Evaluation Strategy

Standard accuracy is misleading in highly imbalanced datasets (e.g., a model that predicts "non-fraud" 100% of the time would have an accuracy of 99.828%). Therefore, we will focus on metrics that prioritize the correct identification of the minority class (fraud):

| Metric               | Calculation (Conceptual)                            | Importance for Fraud Detection                                                                                                                                                   |
|----------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recall (Sensitivity) | True Positives / (True Positives + False Negatives) | This is the most critical business metric. It measures the fraction of actual fraud that was correctly detected. Minimizing <b>False Negatives</b> (missed fraud) is key.        |
| Precision            | True Positives / (True Positives + False Positives) | Measures the fraction of transactions predicted as fraud that were actually fraud. Minimizing <b>False Positives</b> (false alarms) prevents unnecessary review costs.           |
| F1-Score             | 2 × (Precision × Recall / Precision + Recall)       | Our Primary Evaluation Metric. It provides a balanced measure, ensuring the model performs well on both the actual detection rate (Recall) and the false alarm rate (Precision). |

|                                               |                                                      |                                                                                                                                                 |
|-----------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Area Under the Precision-Recall Curve (AUPRC) | Measures the trade-off between Precision and Recall. | Recommended by the dataset authors for highly skewed data, it provides a more informative measure than the traditional AUC-ROC in this context. |
|-----------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

C. Learning Objectives

Through this project, the team (Sathya and Surenther) aims to achieve proficiency in:

- 1. **Imbalanced Data Handling:** Mastering several sampling and modeling techniques essential for real-world rare event prediction.
- 2. **Model Tuning:** Gaining expertise in hyperparameter optimization specific to maximizing the F1 - Score.
- 3. **Risk Interpretation:** Translating abstract performance metrics like Recall and Precision into business-relevant interpretations of financial risk.

3. Risk Assessment and Contingency Planning

A. Risks and Ethical Implications

| Category            | Specific Risk/Implication                                                                                                                                | Mitigation/Assessment Plan                                                                                                                                                                                                                                     |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Interpretation | <b>The</b> PCA-transformed features (V1-V28) lack a direct business context, which limits the ability to explain <i>why</i> a transaction is fraudulent. | Our focus will shift from direct feature interpretation to demonstrating high <b>predictive power</b> and statistical reliability across the entire dataset.                                                                                                   |
| Overfitting         | Extreme class imbalance can cause models to overfit to the small number of fraud cases or rely too heavily on the majority class.                        | Strict use of k-fold cross-validation, especially stratified cross-validation, is essential to ensure that the test folds maintain the original class ratio.                                                                                                   |
| Ethical Risk        | A real-world system could assign higher risk scores to specific groups if unanonymized, biased data were used.                                           | Given that the dataset is already anonymized and lacks demographic features, the ethical risk is low; however, we will ensure that our modeling choices do not introduce spurious correlations with features like Amount, which could lead to unfair outcomes. |

## B. Contingency Plan

If the main approach runs into a significant obstacle—such as consistently weak results across all classification models even after thorough tuning and resampling—the backup plan will go into effect.

### **First Pivot (Modeling):**

If ensemble models like Random Forest or XGBoost still fall short of an acceptable F1-score (for example, anything under 0.6), the focus will shift to an anomaly-detection strategy. In this scenario, fraudulent transactions will be treated as rare events, and methods such as Isolation Forest or One-Class SVM will be explored as alternative modeling paths.

### **Major Pivot (Dataset):**

If the dataset itself proves too complex to produce meaningful predictions, the project will transition to a time-series forecasting problem using a more straightforward public dataset—something along the lines of airline passenger numbers or energy-usage data. This shift will still allow us to demonstrate key predictive-modeling skills, but through techniques like ARIMA or Prophet, which are better suited for structured time-based data.

## 4. Group Logistics and Next Steps

Our team is organized and ready to move forward with the outlined plan.

**Group Members:** Sathya and Surenter

**Communication:** We'll use Microsoft Teams for quick check-ins and file sharing, along with in-person meetings three times a week for more discussions since we're close enough to meet regularly.

**Weekly Meeting Times:** Mondays, Wednesdays, and Fridays at 6:00 PM EST (either in person or on Teams).

The next step is to download the dataset, set up our working environment in Python/Jupyter, and begin Phase 2. This phase will focus on exploring the data, cleaning it, and getting everything ready for the modeling work.